



**Verified Carbon
Standard**

QINGDAO LONGXIN WIND POWER PROJECT PHASE I

JUNO.

Document Prepared by Juno Carbon Investment & Environmental Technology
(Beijing) Co., Ltd.

Project Title	Qingdao Longxin Wind Power Project Phase I
Version	02
Report ID	-
Date of Issue	30-August-2022
Project ID	969
Monitoring Period	28-September-2012 to 20-December-2017
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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The Project installs 24 wind turbines with capacity of 2000kW each and 1 wind turbine with capacity of 1800kW, providing a total installed capacity of 49.8MW to generate clean renewable electricity.

The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing electricity generated by NCPG which is dominated by fossil fuel fired power plants. The estimated annual net electricity generation and GHG emission reductions in the first crediting period (from 08/11/2011 to 07/11/2021) are 99,026MWh and 92,183tCO₂e respectively.

The scenario existing prior to the start of the implementation of the project is: The same electricity output by the project activity would have otherwise been generated by the operation of NCPG connected power plants and by the addition of new generation sources. That is the same as the baseline scenario.

The construction of the Project started on 26/05/2011 and The Project was put into operation on 08/11/2011.

The project has been registered as VCS project with Ref. VCS 969 and VCU of 82,302 tCO₂e have been issued for the monitoring period from 08/11/2011- 27/09/2012. The Project has been registered as a CDM project on 28/09/2012¹ (UNFCCC registration reference number: 6697). The emission reductions of the project have not been issued as CERs.

During this monitoring period from 28/09/2012 to 20/12/2017, the total emission reduction achieved is 472,697tCO₂e.

1.2 Sectoral Scope and Project Type

This category of the Project would fall within sectoral Scope 1: Energy Industry (Renewable/non-renewable sources).

The type of the Project is a wind power generation project.

The project is not a grouped project.

1.3 Project Proponent

Organization name	China Resources Wind Power (Qingdao) Co., Ltd.
Contact person	Zhao Jie
Title	Project Manager

¹ <http://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view>

Address	4th floor, No. 24th floor, No. 217, Renmin Road, Liyuan Office, Pingdu City, Shandong Province, P. R. China.
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Email	zhaojie192@crpower.com.cn

1.4 Other Entities Involved in the Project

Organization name	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.
Role in the Project	VCUs buyer
Contact person	Ding Xue
Title	Technical Director
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1.5 Project Start Date

08/11/2011(commisioning start date)

1.6 Project Crediting Period

08/11/2011-07/11/2021 with a total number of 10 years.

In the registered VCS PD, the crediting period is described as from 08/11/2011 to 27/09/2012. A deviation is requested for the crediting period in the registered VCS PD. The project was registered under VCS Standard Version 3 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project should be updated from 08/11/2011 - 27/09/2012 to 08/11/2011-07/11/2021.

Moreover, the project is also a registered CDM project (Ref. 6697) with a 7*3 renewable project crediting period starting on 28/09/2012, the VCS issuance is not eligible for beyond the end of those 21 years, and the project's expected operation lifetime is 20 years in the registered CDM-PDD. Thus, given that the VCS crediting period started on 08/11/2011, the project proponent confirms the VCS crediting period of the project activity will end on 07/11/2031.

This monitoring period is from 28/09/2012 to 20/12/2017 within the first VCS crediting period and the monitoring period covers 1910 days.

1.7 Project Location

The project is located at Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P. R. China. The nearest Liyuan Town is about 10 kilometres from the project site. The geographical coordinates of the project is from 119.8827°E to 119.9503°E, and from 36.8157°N to 37.8915°N (The central GPS coordinate is 119.91125°E, 36.8568°N). Figure 1 and 2 show the geographical location of the project.



Figure 1. The project on the map of P. R. China



Figure 2. The project on the map of Pingdu City, Shandong Province, P. R. China

1.8 Title and Reference of Methodology

The approved consolidated baseline and monitoring methodology ACM0002: “Consolidated methodology for grid-connected electricity generation from renewable sources” (Version 12.2.0)

Tool for the Demonstration and Assessment of Additionality (Version 05.2.1)

Tool to calculate the emission factor for an electricity system (Version 02.2.0)

For detailed information please refer to:

<https://cdm.unfccc.int/methodologies/index.html>

1.9 Participation under other GHG Programs

The project has been registered as CDM project and the reference number is 6697, the first CDM crediting period is from 28/09/2012 to 27/09/2019. The emission reductions of the project have not been issued as CERs.

The project has been registered as VCS project with Ref. VCS 969 and VCU of 82,302tCO₂e have been issued for the monitoring period from 08/11/2011- 27/09/2012.

1.10 Other Forms of Credit

● Emission Trading Programs and Other Binding Limits

The project has been registered as VCS project with Ref. VCS 969 and VCU of 82,302tCO₂e have been issued for the monitoring period from 08/11/2011- 27/09/2012. All credits from 28/09/2012 to 20/12/2017 will be only claimed under VCS program as VCUs for the project to avoid double counting, which will not be used for compliance with emission trading programs or to meet binding limits on GHG emissions.

The project proponent China Resources Wind Power (Qingdao) Co., Ltd. is an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China ETS. Therefore, the project is not involved in an emission trading program or any other mechanism that includes GHG allowance trading. Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China.

Furthermore, as declared by the project proponent the credits in this monitoring period (28/09/2012- 20/12/2017) has not been counted and will not be counted under other kind of GHG programs.

● Other Forms of Environmental Credit

The project hasn't sought or received another form of environmental credits.

1.11 Sustainable Development Contributions

The project has contributed to sustainable development mainly by:

During this monitoring period, 507,789.778 MWh of electricity from renewable sources has been exported to NCPG, which contributes to SDGs Target 7 “Affordable and Clean Energy” and China's

National Plan on Implementation of the 2030 Agenda for Sustainable Development² 7 through increasing the share of clean energy consumption.

The project creates direct and indirect employment opportunities during construction and operation phases, so it contributes to SDGs Target 8 “Decent Work and Economic Growth” and China’s Implementation Action Plan of “Increasing labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment”. During this monitoring period, 22 people including 6 females were employed by the project owner to responsible for the operation, monitoring and equipment maintenance of this project.

The project produces clean renewable energy by diminishing CO₂ emissions. Therefore, it contributes to SDGs Target 13 “Climate Action” and China’s stated sustainable development priorities “Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas”. The project has supplied 507,789.778 MWh clean energy to NCPG and avoided 472,697 tCO_{2e} release into the atmosphere during this monitoring period.

Refer to table 1 for detail. The evidences of the project’s SD contributions have been provided in the Appendix of this report.

2 <http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf>

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project has supplied 507,789.778 MWh renewable energy to NCPG during this monitoring period.	The project has supplied 596,207.338 MWh renewable energy to NCPG during the first monitoring period and this monitoring period. And 1,980,520 MWh electricity from renewable source are expected to supply to NCPG over the project lifetime which contribute the SDG 7.
2)	8.3	8.3.1 Proportion of informal employment in non-agriculture employment, by sex	Implemented activities to increase	During this monitoring period, 22 people including 6 females were employed by the project owner to responsible for the operation, monitoring and equipment maintenance of this project.	The project activity had provided temporary working opportunities during the construction period. From the operation start date, 22 people were employed including 6 females by the end of this monitoring period.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	The project has supplied 507,789.778 MWh clean energy to NCPG and avoided 472,697 tCO ₂ e release into the atmosphere during this monitoring period.	Prevented the release of 554,999 tCO ₂ e into the atmosphere during the first VCS monitoring period and this monitoring period. And 1,740,680 tCO ₂ e are expected to avoid release into the atmosphere over the project lifetime which contribute the SDG 13.

2 SAFEGUARDS

2.1 No Net Harm

The environmental impact assessment for this project was completed in September 2010 and approved by Qingdao Environmental Protection Bureau on 12th October, 2010. The approval number is Qinghuanshen [2010] 209. A summary of the report is illustrated as below:

Environmental impacts

Ambient air

The impact on ambient air quality of the project is mainly from dust during construction stage, by sprinkling water frequently and timely clearing can reduce the dust pollution. When the project is in operational period, there is no air pollutions. In conclusion, the project will not pose any threat on the quality of ambient air.

Impact from noise

There is some noise during the operation of wind turbines. The equipments and techniques with lower noise have been chosen to apply. Improvement on construction process and strengthening of equipment maintenance is emphasized. The Standard of noise limits of construction site (GB12523-90) is fulfilled during the construction period and Emission standard for industrial enterprise noise at boundary (GB12348-2008) Level II noise standard is fulfilled during operational period. Consequently, the noise of operation has little impact to the surrounding environment. Hence, the noise does not impact the work and daily life of local residents.

Electromagnetic impact

The operation of the wind farm generates electromagnetic pollution, whereas the pollution is slight. In addition, the wind turbines are very far from local residents and village. Therefore, the electronic magnetic pollution to the surrounding environment is insignificant.

Impact from Solid waste

Solid wastes generated from the project activity are excavated earth material and municipal solid waste. Part of the excavated earth material is backfilled, and the rest is used for land levelling and road construction near the project site. The municipal solid waste is collected and treated together with the waste from local residents. As the report indicates, solid waste is handled properly.

Impact from Wastewater

Wastewater is mainly domestic wastewater. Wastewater quantity is fairly small and treatment methods are applied for on-site primary treatment, and then the wastewater is treated together with the local wastewater. Oxidation pond has been built on the site, through which the discharging wastewater is fulfilled the Reuse of recycling water for urban and Water Quality standard for Urban Miscellaneous Water Consumption (GB/T 18920-2002) is used as virescence water. Therefore, the impact of wastewater is limited and mitigated.

Ecological impact

The land where the project located are wasteland and no immigration is involved in the project. Land compensation has carried out according to the relative national regulations. No migrating

birds have been found in the project field. Therefore, the project is not located on the passage of migrating birds, and the project construction has not influence the migration of birds.

During the construction period, some vegetation will be affected, but this will be restored after the construction is completed. The project will be constructed in according with the soil and water conservation plan, which can effectively control soil erosion during the construction period and reduce the impact on the ecological environment.

Socio-economic impacts

As the project is a wind power plant, the project will generate electricity from renewable sources which can reduce the consumption of fossil fuel and GHG emission. Furthermore, the project provides long-term opportunities for local residents and all the employees will be regularly trained with the wind farm operation technology, which contributes to local economic and social sustainable development.

In conclusion, construction and operation of the project does not cause any significant negative environmental nor socio-economic impacts.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

On 20th Mar 2011, the project owner carried out a survey of the local residents at Zhujiaying Village, Shuangmiao Village and Zhuangjiaying Village which belong to Liyuan Town and are around the project location. The staff introduced the background of the project and then sent out 50 copies of questionnaire in a random way, 49 pieces of reply were received. Among the interviewees, 22 of them are farmers, 4 are workers, 13 are governmental officials, 7 are students and 3 others.

The questions regarding the project were mainly as follows:

1. How is the current environment quality?
2. Are there any negative impacts of the proposed project on the everyday life of local residents?
3. Is the proposed project going to help improve the local economy?
4. Does the proposed project impact on local sound/noise environment?
5. Which is the environmental topic that concerns you the most during the construction and operation of the proposed project?
6. Do you support the proposed project?

The summary of questionnaire survey is listed as the following:

- 37 (76%) of them think the current environment quality is good, 10 (20%) think it is normal and 2 (4%) of them are unsure;
- 38 (79%) of them think there will not be any negative impacts on their everyday life, 11 (27%) of them are not sure;

- 36 (73%) of them think the project will help to improve the local economy, 13(27%) of them are not sure;
- 20 (41%) of them think the project doesn't impact on local sound/noise environment, 5 (10%) of them think it will, 27 (49%) of them are unsure;
- Regarding the construction and operation of the project, 27 (55%) of them are most concerned with electromagnetic interference, 19 (40%) of them are most concerned with the noise level and 3 (6%) of them are most concerned with wastewater;
- 45 (92%) of them support the implementation of the project and 4 (8%) of them don't care.

The summary of local stakeholders meeting: The local community possesses basically positive comments on the effects of the project. The interviewees considered that local social, economic and environmental development would be beneficial from the project. The response was overall supportive to the project implementation.

LSC during the operation period

In order to keep on-going communications with local stakeholders, the project owner has public its phone number of the central control room at the power plant gate and oral notice to local people and village committees. Anyone who have comments on the project could phone the project owner directly. Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations.

There are no negative comments received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is not AFOLU project.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

This project installs 24 wind turbines with capacity of 2000 kW each and 1 wind turbines with capacity of 1800 kW which amount to a total installed capacity of 49.8MW. The estimated power output is 99,026MWh/year and the load factor is 22.70%. The Project started operation on 08/11/2011.

The main technical specifications of the wind turbine are provided in the following table.

Item	Unit	Index	Index
Model	-	V90-2.0MW	V90-1.8MW
Rated capacity	kW	2000	1800
Number of blades	-	3	3
Rotor diameter	M	90	90

Rated wind speed	m/s	13	12
Shutoff speed	m/s	2.5	3.5
Height of hub	m	80	80
Rated	V	690	690

An 110kV substation is built at the project site, which connects the project to the Tangtian 220kV Substation which is about 6.7 km away from the project and belongs to the North China Power Grid. Then the power produced in the wind farm can be transmitted to the North China Power Grid.

Bidirectional meter M1 with accuracy 0.5S was installed on the Tangtian Substation to measure the total electricity supplied by the Project and Phase II project, meanwhile M1 is used to measure the total electricity imported from the grid by the Project and Phase II project. Meter M2 and M3 installed at the on-site substation recorded the electricity supplied by the Project and Phase II Project respectively.

Meter M4 is used to measure the electricity imported from the grid by the project and Phase II project via a spare 10kV back-up line.

During this monitoring period, the Project had been in normal operation, without special events such as overhaul or exchange of equipment. There were no events or situations which may impact the applicability of the methodology.

The monitoring system was implemented in line with the monitoring plan in the registered PDD. The actual operation was conducted strictly in compliance with the description in the monitoring plan.

3.2 Deviations

3.1.1 Methodology Deviations

There is no methodology deviation applied during this monitoring period.

3.1.2 Project Description Deviations

In the registered VCS PD, the crediting period is described as from 08/11/2011 to 27/09/2012. A deviation is requested for the crediting period in the registered VCS PD. The project is registered under VCS Standard Version 3 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project should be updated from 08/11/2011 - 27/09/2012 to 08/11/2011-07/11/2021. Moreover, the project is also a registered CDM project (Ref. 6697) with a 7*3 renewable project crediting period starting on 28/09/2012, the VCS issuance is not eligible for beyond the end of those 21 years, and the project's expected operation lifetime is 20 years in the registered CDM-PDD. Thus, given that the VCS crediting period started on 08/11/2011, the project proponent confirms the total VCS crediting period of the project activity will end on 07/11/2031.

The deviation on the crediting period has no impacts on the applicability, additionality or baseline of the project. There is no other project description deviation during this monitoring period.

3.3 Grouped Projects

The project is not a grouped project.

4 DATA AND PARAMETERS

1.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ e/MWh
Description	Baseline emission factor of NCPG in year y.
Source of data	The registered PDD.
Value applied	0.9309
Justification of choice of data or description of measurement methods and procedures applied	Ex-ante determined in the registered PD and fixed during the 1 st crediting period.
Purpose of Data	Calculation of baseline emissions.
Comments	/

1.2 Data and Parameters Monitored

Data / Parameter	EG _{output,y}
Data unit	MWh
Description	Total electricity supplied to the Grid by the Project and Phase II project
Source of data	Measured by the electricity meter M1
Description of measurement methods and procedures to be applied	Bidirectional meter M1 with the accuracy of 0.5S installed on the Tangtian Substation is owned by the grid company to measure the electricity supplied by the Project and Phase II project. The meter M1 measured the electricity continuously, the readings are daily recorded by the project owner and the grid company, the monthly cut-off time is determined by the project owner and the grid company in advance every month³.
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated
Value monitored	1,075,354.250

³ During this monitoring period, the cut-off date to record the meter are mainly day 23, 24, 25 and 26 every month, and the specific time of meter reading is determined by the project owner and the grid company in advance every month.

Monitoring equipment	M1 is a bidirectional meter. The bidirectional meter M1 at Tangtian substation was exchanged on 18/10/2016 according to the agreement of the project owner and the grid company. The information of the new and old meter is as follows:																
		<table><tr><td></td><td>The old meter</td><td>The new meter</td></tr><tr><td>Type</td><td>DTZ341</td><td>DTZ666</td></tr><tr><td>Accuracy class</td><td>0.5S</td><td>0.5S</td></tr><tr><td>Serial number</td><td>QD0243006231</td><td>3730001000000188595727</td></tr><tr><td>Calibration frequency</td><td>at least annually</td><td>at least annually</td></tr></table>		The old meter	The new meter	Type	DTZ341	DTZ666	Accuracy class	0.5S	0.5S	Serial number	QD0243006231	3730001000000188595727	Calibration frequency	at least annually	at least annually
		The old meter	The new meter														
	Type	DTZ341	DTZ666														
	Accuracy class	0.5S	0.5S														
	Serial number	QD0243006231	3730001000000188595727														
Calibration frequency	at least annually	at least annually															
QA/QC procedures to be applied																	
The accuracy of meter M1 is 0.5S, which is consistent with the monitoring plan and the Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). The reading records have been cross-checked with the sales receipts every month. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.																	
Purpose of the data	Calculation of baseline emissions																
Calculation method	N/A																
Comments	/																

Data / Parameter	EG _{another,y}
Data unit	MWh
Description	Electricity generated by the Phase II project
Source of data	Measured by the electricity meter M3
Description of measurement methods and procedures to be applied	Bidirectional meter M3 with the accuracy of 0.5S installed on the 110kV outline of the main transformer of Phase II project on-site is used to measure the electricity generated by the Phase II project. The meter M3 measured the electricity continuously, the readings are daily recorded and monthly aggregated by the project owner.
Frequency of monitoring/recording	Continuously measured, Daily recorded and monthly aggregated

Value monitored	563,813.747
Monitoring equipment	M3 is a bidirectional meter. Type: DTSD341 Accuracy class: 0.5S Serial number: 11060727990140 Calibration frequency: at least annually
QA/QC procedures to be applied	The accuracy of meter M3 is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Purpose of the data	Calculation of baseline emissions
Calculation method	N/A
Comments	/

Data / Parameter	EG _{import,y}
Data unit	MWh
Description	Total electricity imported from the Grid by the Project and Phase II project through the 110 KV line
Source of data	Measured by the electricity meter M1
Description of measurement methods and procedures to be applied	Bidirectional meter M1 with the accuracy of 0.5S installed on the Tangtian Substation is used to measure the electricity imported from the Grid by the Project and Phase II project. The meter M1 measured the electricity continuously, the readings are daily recorded by the project owner and the grid company, the monthly cut-off time is determined by the project owner and the grid company in advance every month.
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated
Value monitored	3,647.124
Monitoring equipment	M1 is a bidirectional meter. The bidirectional meter M1 at Tangtian substation was exchanged on 18/10/2016 according to the agreement of the project owner

	and the grid company. The information of the new and old meter is as follows:		
		The old meter	The new meter
	Type	DTZ341	DTZ666
	Accuracy class	0.5S	0.5S
	Serial number	QD0243006231	3730001000000188 595727
	Calibration frequency	at least annually	at least annually
QA/QC procedures to be applied	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). The reading records have been cross-checked with the sales receipts every month. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.		
Purpose of the data	Calculation of baseline Emission		
Calculation method	N/A		
Comments	/		

Data / Parameter	EG _{im-spare,y}
Data unit	MWh
Description	Electricity imported from the Grid by the project and Phase II Project through a spare 10kV line
Source of data	Measured by the electricity meter M4
Description of measurement methods and procedures to be applied	Meter M4 with the accuracy of 1.0 installed on a spare 10kV line is used to measure the electricity imported from the grid by the Project and Phase II project. The meter M4 measured the electricity continuously, and the readings are monthly recorded by the project owner and the grid company together on the Appointed date.
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated
Value monitored	103.600

Monitoring equipment	The meter M4 installed on a spare 10kV line was exchanged on 11/10/2015 according to the agreement of the project owner and the grid company. The information of the new and old meter is as follows: <table><tr><td></td><td>The old meter</td><td>The new meter</td></tr><tr><td>Type</td><td>DTZ522</td><td>DTZ522</td></tr><tr><td>Accuracy class</td><td>1.0</td><td>1.0</td></tr><tr><td>Serial number</td><td>DT4803509</td><td>3700100423373 25</td></tr><tr><td>Calibration frequency</td><td>at least annually</td><td>at least annually</td></tr></table>				The old meter	The new meter	Type	DTZ522	DTZ522	Accuracy class	1.0	1.0	Serial number	DT4803509	3700100423373 25	Calibration frequency	at least annually	at least annually
	The old meter	The new meter																
Type	DTZ522	DTZ522																
Accuracy class	1.0	1.0																
Serial number	DT4803509	3700100423373 25																
Calibration frequency	at least annually	at least annually																
QA/QC procedures to be applied	The accuracy of meters is 1.0, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). The reading records have been cross-checked with the sales receipts every month. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.																	
Purpose of the data	Calculation of baseline Emission																	
Calculation method	N/A																	
Comments	/																	

Data / Parameter	EG _{facility,y}
Data unit	MWh
Description	Net electricity supplied to the Grid by the project
Source of data	Readings from electricity meter M1, M3 and M4.
Description of measurement methods and procedures to be applied	Bidirectional meter M1 with the accuracy of 0.5S installed on the Tangtian Substation is used to measure the electricity supplied to the grid by the Project and Phase II project (EG_{output,y}) and electricity imported from the grid by the Project and Phase II project through the 110kV line (EG_{import,y}). Bidirectional meter M3 with the accuracy of 0.5S installed on the 110kV outline of the main transformer of Phase II project on-site is used to measure the electricity generated by the Phase II project.

	Meter M4 with the accuracy of 1.0 installed on a spare 10kV line is used to measure the electricity imported from the grid by the Project and Phase II project.		
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated		
Value monitored	507,789.778		
Monitoring equipment	M1 is a bidirectional meter.		
		The old meter	The new meter
	Type	DTZ341	DTZ666
	Accuracy class	0.5S	0.5S
	Serial number	QD0243006231	373000100000018 8595727
	Calibration frequency	at least annually	at least annually
	M3 is a directional meter.		
	Type: DTSD341		
	Accuracy class: 0.5S		
	Serial number: 11060727990140		
	Calibration frequency: at least annually		
	M4 is a unidirectional meter.		
		The old meter	The new meter
	Type	DTZ522	DTZ522
	Accuracy class	1.0	1.0
Serial number	DT4803509	370010042337325	
Calibration frequency	at least annually	at least annually	
QA/QC procedures to be applied	The accuracy of M1, M3 and M4 is 0.5S, 0.5S and 1.0 respectively, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 or DL/T448-2016). Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.		
Purpose of the data	Calculation of baseline Emission		

Calculation method	The net electricity supplied to the Grid by the project were calculated by the following formula: $EG_{\text{facility},y} = EG_{\text{output},y} - EG_{\text{another},y} - EG_{\text{import},y} - EG_{\text{im-spare},y}$
Comments	/

Table 2 Main information of the electricity meters

Meters	Type	Serial No.	Accuracy	Calibration date	Valid Date	Validity of Certifier
M1 ⁴	DTZ341 (The old meter)	QD024300 6231	0.5S	20/09/2011	19/09/2012	Shandong Institute of Metrology
				18/09/2012	17/09/2013	
				09/04/2013	08/04/2014	
				08/04/2014	07/04/2015	State Grid Shandong Electric Power Metering Center
				06/04/2015	05/04/2016	
				05/04/2016	18/10/2016	
M2	DTZ666 (The new meter)	37300010 00000188 595727	0.5S	09/10/2016	08/10/2017	State Grid Shandong Electric Power Metering Center
				03/04/2017	02/04/2018	
				14/10/2011	13/10/2012	Shandong Institute of Metrology
				18/09/2012	17/09/2013	
				09/04/2013	08/04/2014	
M3	DTSD341	11060727 990139	0.5S	08/04/2014	07/04/2015	State Grid Shandong Electric Power Metering Center
				06/04/2015	05/04/2016	
				05/04/2016	04/04/2017	
				03/04/2017	02/04/2018	Shandong Institute of Metrology
				10/04/2012	09/04/2013	
				09/04/2013	08/04/2014	
M4 ⁵	DTSD341	11060727 990140	0.5S	08/04/2014	07/04/2015	State Grid Shandong Electric Power Metering Center
				06/04/2015	05/04/2016	
				05/04/2016	04/04/2017	
				03/04/2017	02/04/2018	Shandong Institute of Metrology
				14/10/2011	13/10/2012	
				18/09/2012	17/09/2013	
M5	DTZ522 (The old meter)	DT480350 9	1.0	09/04/2013	08/04/2014	Shandong Institute of Metrology
				08/04/2014	07/04/2015	
				06/04/2015	10/10/2015	
	DTZ522 (The new meter)	37001004 2337325	1.0	10/08/2015	09/08/2016	State Grid Shandong Electric Power Metering Center
				05/04/2016	04/04/2017	
				03/04/2017	02/04/2018	

⁴ The bidirectional meter M1 at Tangtian substation was exchanged on 18/10/2016.

⁵ The meter M4 installed on a spare 10kV line was exchanged on 11/10/2015.

1.3 Monitoring Plan

1. Monitoring equipment and data collection procedures

The electric energy metering system was equipped according to the requirements of the Technical administrative code of electric energy metering (DL/T448-2020 or DL/T448-2016).

The electricity supplied by the Project to the Grid is transmitted from the on-site substation of the Project through the 110kV transmission line and then be delivered to the Grid and share the 110kV transmission line with the Phase II Project, which is developed by the same project owner with the Project.

Bidirectional meter M1 with accuracy class of 0.5S installed at the Tangtian Substation is used to measure the total electricity supplied to the grid by the Project and Phase II project ($EG_{\text{output},y}$) and the electricity imported from the grid by the project and Phase II project through the 110kV line ($EG_{\text{import},y}$). Grid company is responsible for the operation of M1, the meter M1 measured the electricity continuously, the readings are daily recorded by the project owner and the grid company, the monthly cut-off time is determined by the project owner and the grid company in advance every month⁶. The reading records will be cross-checked with the sales receipts every month.

Meter M2 with accuracy class of 0.5S installed at the 110kV outline of the main transformer of the project on-site is used to measure the electricity supplied to the grid by the Project. The reading of meter M2 is carried out by the project owner for internal quality control of the project which is not relating to the monitoring of any parameter to be monitored required by the monitoring plan.

Meter M3 with accuracy class of 0.5S installed at the 110kV outline of the main transformer of Phase II project on-site is used to measure the electricity generated by the Phase II project ($EG_{\text{another},y}$). The reading of meter M3 is recorded daily by the project owner and aggregated monthly.

Additionally, a spare 10kV line is used for supplying electricity in order to start-up or emergencies. Electricity imported from the grid by the project and Phase II project via this spare 10kV line ($EG_{\text{im-spare},y}$) is be measured by meter M4 with the accuracy of 1.0. The reading of meter M4 is recorded daily by the project owner and aggregated monthly and after confirmed by both sided, the sales receipts were issued for cross-check.

During this monitoring period, the cut-off date to record the meter are mainly day 23, 24, 25 and 26 every month, and the specific time of meter reading is determined by the project owner and the grid company in advance every month.

⁶ During this monitoring period, the cut-off date to record the meter are mainly day 23, 24, 25 and 26 every month, and the specific time of meter reading is determined by the project owner and the grid company in advance every month.

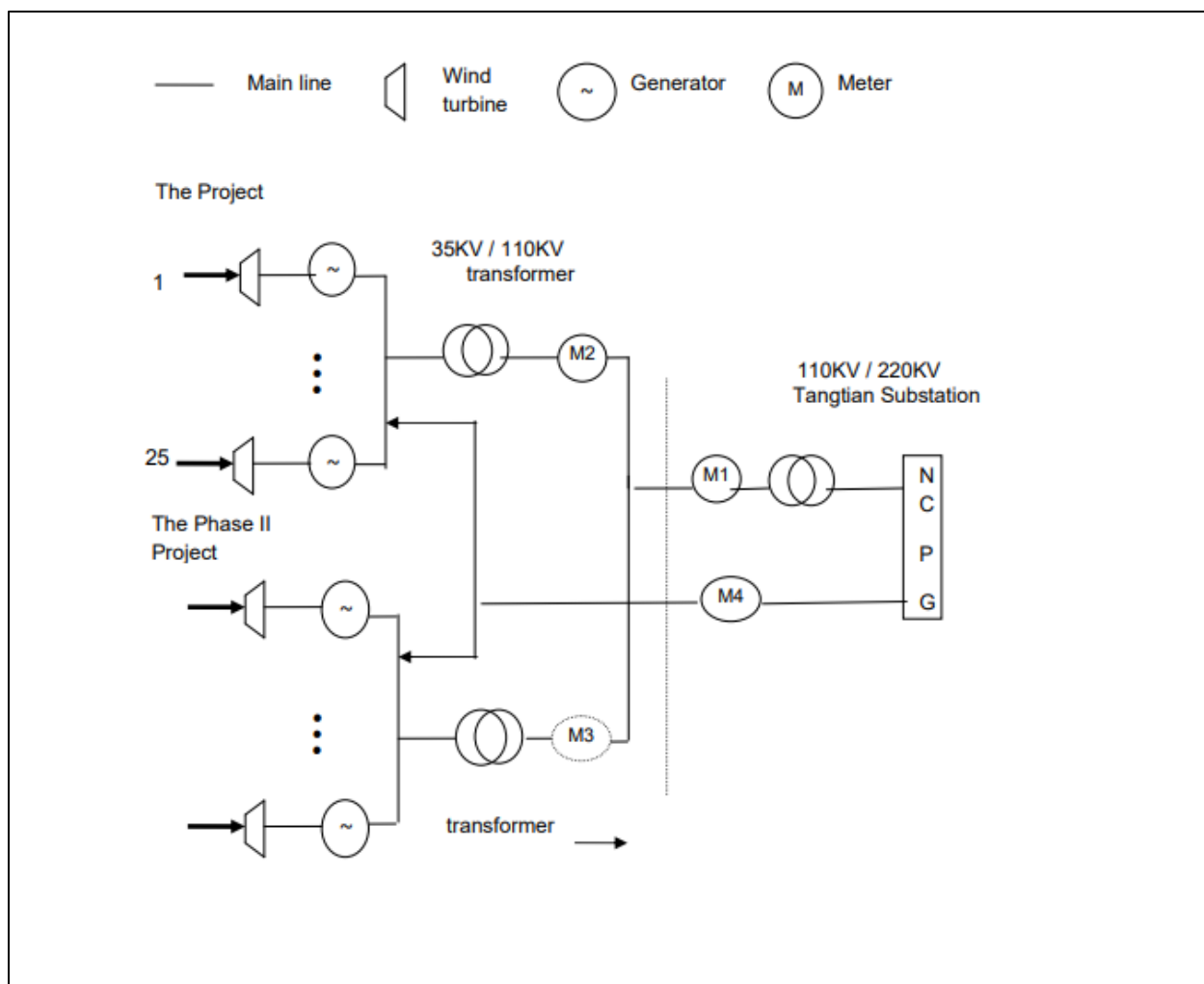


Figure 3. The flowchart of the Project

The net electricity supplied to the grid by the Project is calculated as followed:

$$EG_{\text{facility},y} = EG_{\text{output},y} - EG_{\text{another},y} - EG_{\text{import},y} - EG_{\text{im-spares},y}$$

The project owner keeps the records of the electricity meter's readings and sales receipts for verification by the VVB.

2. Organizational structure

China Resources Wind Power (Qingdao) Co., Ltd., the project owner has used this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan. This plan should be modified according to actual conditions and requirements of VVB in order to ensure that the monitoring is credible, transparent, and conservative.

The responsibilities of the project staff are as follow:

The General Manager of the project entity will appoint a VCS project manager. The Operational and Monitoring Manager of the plant, the Financial Chief, and the Technical Chief are responsible for the collection of the data and information required in the monitoring plan.

General Manager: To be responsible for supervising the whole monitoring procedure.

VCS Project Manager: To be responsible for data management and compiling monitoring report.

Operational and monitoring manager: To be responsible for collecting data and do internal audit.

Financial chief: To be responsible for collection of sales receipts.

Technical chief: To be responsible for preparing operational reports of the project activity, recording the daily operation of the wind farm, including operating periods, equipment defects, etc.

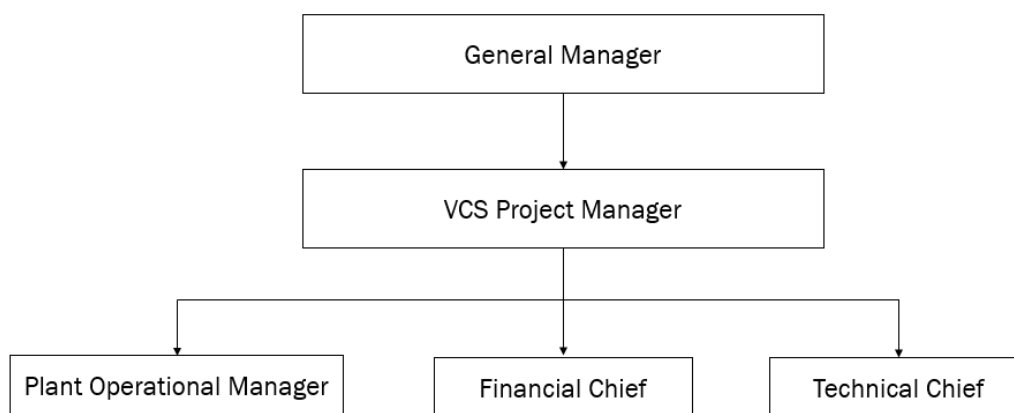


Figure 4. Management Structure of Monitoring Plan

3. Emergency procedures

In case metering equipment is damaged and no reliable readings can be recorded the project entity will estimate net supply by the project activity according to the following procedure:

In case metering equipment is damaged: The project entity and the grid company will jointly calculate a conservative estimate of power supplied to the grid. A statement will be prepared indicating

I: the background to the damage to metering equipment;

II: the assumptions used to estimate net supply to the grid for the days for which no record could be recorded;

III: the estimation of power supplied to the grid. The statement will be signed by both a representative of the project entity as well as a representative of the grid company.

4. Quality Assurance and Quality Control

The metering equipment will be properly calibrated and checked according to the rules of Relative Technical Administrative Code of Electric Energy Metering (DL/T448-2000 and DL/T448-2016). The project owner will prepare backup procedures to deal with any errors occurred to the meters. The calibration records carried out by the grid company should be provided to the project owner, and these records will be maintained by the project owner and the third party designated.

Meters should be tested by a qualified metric organization co-authorized by the project owner and the grid company within 10 days after: the equipments malfunction caused by improper operation All the calibration test records should be maintained safely for the verification.

5. Monitoring Training

Monitoring training is critical to ensure that all members of the Monitoring Team have a thorough understanding of the monitoring procedure and are able to carry out the monitoring tasks strictly in line with the monitoring plan. The team leader is responsible for evaluating training outcome. Only qualified staffs can work on duty.

The training will include:

I: Training on operation and monitoring system of the wind farm. This is the type of training which are routinely carried out by the wind farm itself for new staffs.

II: Training on CDM basics with focus on monitoring. It will be carried out by the CDM consultancy company before the project is implemented.

The CDM monitoring manual will be used as the primary training materials. The training includes the following contents:

- CDM project cycle and the significance of monitoring
- Management structure and work scope of each team member
- Components of the monitoring plan
- QA/QC procedure
- Monitoring report template
- Preparation for verification

6. Data management

The project owners keep the records of the data collected during monitoring, including but not limited to the meter readings of the meters, copies of sales receipts and the operation log etc. After the technical chief prepare the monitoring data every monthly, the plant operational and monitoring manger cross checked the electricity with the power generated, and also the meter readings of M2 for internal data quality assurance. If there are any discrepancy happened, the monitoring will be rechecked to see if there is an error in the log or if the meter is faulty. Meanwhile, the financial chief will check the monitoring data with the sales receipts every month. At last, the monitoring data are verified by the Office Manager once every quarter.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions in the monitoring period are calculated by multiplying the fixed baseline emission factor determined in the registered PDD of the project ex ante by net electricity supplied by the Project to the NCPG in the monitoring period.

$$BE_y = EG_{\text{facility},y} \times EF_{\text{grid,CM},y}$$

$$EG_{\text{facility},y} = EG_{\text{output},y} - EG_{\text{another},y} - EG_{\text{import},y} - EG_{\text{im-spare},y}$$

Where:

BE_y : Baseline emissions in year y. (in tCO_2e) ;

$EG_{facility,y}$: Net electricity supplied to the Grid by the project in year y (in MWh) ;

$EF_{grid,CM,y}$: Baseline emission factor of NCPG (in tCO_2e/MWh);

$EG_{output,y}$: Total electricity supplied to the Grid by the project and Phase II project in the year y (in MWh) ;

$EG_{another,y}$: Electricity generated by the Phase II Project in the year y (in MWh);

$EG_{import,y}$: Total electricity imported from the Grid by the project and Phase II project in the year y (in MWh);

$EG_{im-spare,y}$: Electricity imported from the Grid by the project and Phase II project through a spare 10 kV line in the year y (in MWh)

The $EG_{facility,y}$, and BE_y are listed as follow:

$EG_{facility,y} = 507,789.778$ MWh

$BE_y = 472,697$ tCO_2e

Table 3 Net electricity supplied to the Grid by the project

Monitoring period		Total electricity supplied to the Grid by the Project and Phase II Project (EG _{output,y}) unit: MWh			Total electricity imported from the Grid by the project and Phase II project through the 110kV line (EG _{import,y}) unit: MWh			Electricity imported from the Grid by the project and Phase II project through a spare 10 kV line (EG _{im-spare}) unit: MWh			Electricity generated by the Phase II Project (EG _{another,y}) unit: MWh	Net electricity supplied to the grid by the project (EG _{facility}) unit: MWh
From	To	Data based on meter readings M1	Data based on sales receipt	Conservative value	Data based on meter readings M1(110kV)	Data based on sales receipts	Conservative values	Data based on meter readings M4(10kV)	Data based on sales receipts	Conservative values	Data based on meter readings M3	Calculated by EG _{output,y} -EG _{import,y} -EG _{im-spare,y} +EG _{another,y}
		A	B	C=Min (A, B)	D	E	F=Min (D, E)	G	H	I=Max (G, H)	J	J=C-F-I-J
28/09/2012	25/10/2012	15800.810	15800.810	15800.810	39.327	102.004	102.004	0.429	0.429	0.429	8682.080	7,016.297
26/10/2012	25/11/2012	21600.480	21600.480	21600.480	30.360	29.04	30.360	0.461	0.461	0.461	11698.720	9,870.939
26/11/2012	23/12/2012	21933.120	21933.120	21933.120	27.720	35.640	35.640	0.461	0.461	0.461	12515.360	9,381.659
24/12/2012	31/12/2012	4709.760	4709.760	4709.760	7.920	6.93	7.920	0.115	0.115	0.115	2442.880	2,258.845
Subtotal 2012				64044.170			175.924			1.466	35339.040	28,527.740
01/01/2013	25/01/2013	19315.560	19315.560	19315.560	23.760	20.790	23.760	0.346	0.346	0.346	9368.480	9,922.974
26/01/2013	25/02/2013	11820.600	11820.600	11820.600	151.800	149.160	151.800	0.461	0.461	0.461	7084.000	4,584.339
26/02/2013	26/03/2013	26550.480	26550.480	26550.480	15.840	18.480	18.480	0.461	0.461	0.461	14490.080	12,041.459
27/03/2013	24/04/2013	28005.120	28005.120	28005.120	6.600	10.560	10.560	0.461	0.461	0.461	14414.400	13,579.699
25/04/2013	25/05/2013	24845.040	24845.040	24845.040	15.840	14.520	15.840	0.461	0.461	0.461	13221.120	11,607.619
26/05/2013	24/06/2013	19561.080	19561.080	19561.080	21.120	0.000	21.120	0.461	0.461	0.461	10797.600	8,741.899
25/06/2013	25/07/2013	17169.240	17169.240	17169.240	33.000	35.640	35.640	0.461	0.461	0.461	9275.200	7,857.939
26/07/2013	25/08/2013	10152.120	10152.120	10152.120	42.240	44.880	44.880	0.461	0.461	0.461	5427.840	4,678.939
26/08/2013	24/09/2013	11993.520	11993.520	11993.520	56.760	62.040	62.040	2.297	2.297	2.297	6246.240	5,682.943
25/09/2013	25/10/2013	18540.720	18540.720	18540.720	73.920	64.680	73.920	0.461	0.461	0.461	9584.960	8,881.379
26/10/2013	24/11/2013	17376.480	17376.480	17376.480	80.520	73.920	80.520	0.461	0.461	0.461	8780.640	8,514.859
25/11/2013	25/12/2013	21795.840	21795.840	21795.840	43.560	59.400	59.400	0.461	0.461	0.461	11114.400	10,621.579
26/12/2013	31/12/2013	4819.320	4819.320	4819.320	1.320	7.128	7.128	0.092	0.092	0.092	2826.560	1,985.540
Subtotal 2013				231945.120			605.088			7.345	122631.520	108,701.167
01/01/2014	24/01/2014	11274.120	11274.200	11274.120	33.000	28.512	33.000	0.369	0.369	0.369	6052.640	5,188.111
25/01/2014	25/02/2014	17651.040	17651.040	17651.040	40.920	36.960	40.920	0.461	0.461	0.461	9597.280	8,012.379

26/02/2014	25/03/2014	19215.240	19215.240	19215.240	34.320	40.920	40.920	0.461	0.461	0.461	9347.360	9,826.499
26/03/2014	25/04/2014	16573.920	16573.920	16573.920	92.400	68.640	92.400	0.461	0.461	0.461	9285.760	7,195.299
26/04/2014	26/05/2014	28915.920	28915.920	28915.920	35.640	54.120	54.120	0.461	0.461	0.461	14567.520	14,293.819
27/05/2014	24/06/2014	16312.560	16312.560	16312.560	62.040	46.200	62.040	0.461	0.461	0.461	8050.240	8,199.819
25/06/2014	26/07/2014	13211.880	13211.880	13211.880	50.160	71.280	71.280	0.461	0.461	0.461	7143.840	5,996.299
27/07/2014	26/08/2014	7539.840	7539.840	7539.840	91.080	75.240	91.080	0.461	0.461	0.461	3706.560	3,741.739
27/08/2014	25/09/2014	4623.960	4623.960	4623.960	120.120	122.760	122.760	0.461	0.461	0.461	2455.200	2,045.539
26/09/2014	26/10/2014	16060.440	16060.440	16060.440	33.000	44.880	44.880	0.461	0.461	0.461	8608.160	7,406.939
27/10/2014	24/11/2014	14810.400	14810.400	14810.400	81.840	75.240	81.840	0.461	0.461	0.461	7080.480	7,647.619
25/11/2014	25/12/2014	27009.840	27009.840	27009.840	39.600	52.800	52.800	0.461	0.461	0.461	14078.240	12,878.339
26/12/2014	31/12/2014	3033.360	3033.360	3033.360	22.440	11.886	22.440	0.086	0.086	0.086	1561.120	1,449.714
Subtotal 2014				196232.520			810.480			5.526	101534.400	93882.113
01/01/2015	25/01/2015	15265.800	15265.800	15265.800	40.920	47.514	47.514	0.375	0.375	0.375	8210.400	7,007.511
26/01/2015	25/02/2015	13902.240	13902.240	13902.240	68.640	62.040	68.640	0.461	0.461	0.461	7332.160	6,500.979
26/02/2015	25/03/2015	18724.200	18724.200	18724.200	36.960	40.920	40.920	0.461	0.461	0.461	9241.760	9,441.059
26/03/2015	25/04/2015	26071.320	26071.320	26071.320	18.480	23.760	23.760	6.083	6.083	6.083	13759.680	12,281.797
26/04/2015	25/05/2015	21658.560	21658.560	21658.560	30.360	29.040	30.360	0.461	0.461	0.461	11098.560	10,529.179
26/05/2015	25/06/2015	18256.920	18256.920	18256.920	40.920	34.320	40.920	0.461	0.461	0.461	9771.520	8,444.019
26/06/2015	25/07/2015	10775.160	10775.160	10775.160	43.560	51.480	51.480	0.461	0.461	0.461	5624.960	5,098.259
26/07/2015	25/08/2015	8338.440	8338.440	8338.440	89.760	72.600	89.760	3.766	3.766	3.766	4241.600	4,003.314
26/08/2015	25/09/2015	7213.800	7213.800	7213.800	85.800	89.760	89.760	22.177	22.177	22.177	3991.680	3,110.183
26/09/2015	25/10/2015	15859.800	15859.800	15859.800	54.120	52.800	54.120	20.222	20.222	20.222	7835.520	7,949.938
26/10/2015	25/11/2015	17589.000	17589.000	17589.000	59.400	71.280	71.280	8.108	8.108	8.108	9570.880	7,938.732
26/11/2015	26/12/2015	13702.920	13702.920	13702.920	21.120	15.840	21.120	9.858	9.858	9.858	6834.080	6,837.862
27/12/2015	31/12/2015	2566.080	2566.080	2566.080	6.600	11.220	11.220	0.077	0.077	0.077	1092.960	1,461.823
Subtotal 2015				189924.240			640.854			72.971	98605.760	90604.655
01/01/2016	26/01/2016	18068.160	18068.160	18068.160	54.120	56.100	56.100	0.384	0.384	0.384	9292.800	8,718.876
27/01/2016	24/02/2016	17657.640	17657.64	17657.640	40.920	43.560	43.560	0.461	0.461	0.461	9109.760	8,503.859
25/02/2016	26/03/2016	18757.200	18757.20	18757.200	29.040	29.040	29.040	0.461	0.461	0.461	10035.520	8,692.179
27/03/2016	25/04/2016	23687.400	23687.40	23687.400	31.680	29.040	31.680	0.461	0.461	0.461	12242.560	11,412.699
26/04/2016	26/05/2016	23040.600	23040.60	23040.600	30.360	26.400	30.360	0.461	0.461	0.461	11792.000	11,217.779

27/05/2016	24/06/2016	14506.800	14506.80	14506.800	48.840	52.800	52.800	0.461	0.461	0.461	7935.840	6,517.699
25/06/2016	26/07/2016	17466.240	17466.24	17466.240	46.200	50.558	50.558	1.383	1.383	1.383	8986.560	8,427.739
27/07/2016	26/08/2016	7702.200	7702.20	7702.200	91.080	71.280	91.080	0.461	0.461	0.461	4341.920	3,268.739
27/08/2016	26/09/2016	13700.280	13700.28	13700.280	79.200	97.680	97.680	0.461	0.461	0.461	6825.280	6,776.859
27/09/2016	26/10/2016	12980.880	12980.88	12980.880	80.520	71.280	80.520	1.619	1.619	1.619	6696.800	6,201.941
27/10/2016	26/11/2016	17926.920	17926.92	17926.920	52.800	59.400	59.400	0.461	0.461	0.461	9238.240	8,628.819
27/11/2016	26/12/2016	15553.560	15553.56	15553.560	68.640	51.480	68.640	0.461	0.461	0.461	8377.600	7,106.859
27/12/2016	31/12/2016	3340.920	3340.920	3340.920	14.520	16.720	16.720	0.077	0.077	0.077	1344.640	1,979.483
Subtotal 2016				204388.800			708.138			7.612	106219.520	97453.530
01/01/2017	26/01/2017	10350.120	10350.120	10350.120	67.320	83.600	83.600	0.384	0.384	0.384	6691.520	3,574.616
27/01/2017	26/02/2017	24303.840	24303.840	24303.840	48.840	47.520	48.840	0.461	0.461	0.461	11281.600	12,972.939
27/02/2017	26/03/2017	14939.760	14939.760	14939.760	26.400	23.760	26.400	0.461	0.461	0.461	7978.080	6,934.819
27/03/2017	26/04/2017	17109.840	17109.840	17109.840	54.120	54.120	54.120	0.461	0.461	0.461	9074.560	7,980.699
27/04/2017	26/05/2017	23114.520	23114.520	23114.520	43.560	43.560	43.560	0.461	0.461	0.461	11932.800	11,137.699
27/05/2017	20/06/2017	14199.240	14199.240	14199.240	40.920	40.920	40.920	0.461	0.461	0.461	7606.720	6,551.139
21/06/2017	20/07/2017	13824.360	13824.360	13824.360	33.000	33.000	33.000	1.148	1.148	1.148	7307.027	6,483.185
21/07/2017	20/08/2017	11556.600	11556.600	11556.600	102.500	102.960	102.960	0.461	0.461	0.461	5781.600	5,671.579
21/08/2017	20/09/2017	6585.480	6585.480	6585.480	100.320	100.320	100.320	2.999	2.999	2.999	3375.680	3,106.481
21/09/2017	20/10/2017	13152.480	13152.480	13152.480	67.320	67.320	67.320	0.461	0.461	0.461	6716.160	6,368.539
21/10/2017	20/11/2017	20602.560	20602.560	20602.560	64.680	64.680	64.680	0.461	0.461	0.461	11200.640	9,336.779
21/11/2017	20/12/2017	19080.600	19080.600	19080.600	40.920	40.920	40.920	0.461	0.461	0.461	10537.120	8,502.099
Subtotal 2017				188819.400			706.640			8.680	99483.507	88620.573
Total				1075354.250			3647.124			103.600	563813.747	507789.778

Table 4 Calculation of baseline emissions

Monitoring period		Net electricity supplied to the grid by the project (EG _{facility}) unit: MWh	Emission Factor (EF _{grid,CM,y}) unit: tCO _{2e} /MWh	Baseline Emission (BE _y) unit: tCO _{2e}
From	To			
28/09/2012	31/12/2012	28,527.740	0.9309	26,556
01/01/2013	31/12/2013	108,701.167	0.9309	101,189
01/01/2014	31/12/2014	93,882.113	0.9309	87,394
01/01/2015	31/12/2015	90,604.655	0.9309	84,343
01/01/2016	31/12/2016	97,453.530	0.9309	90,719
01/01/2017	20/12/2017	88,620.573	0.9309	82,496
Total		507,789.778	-	472,697

5.2 Project Emissions

The Project is a renewable electricity generating activity. According to ACM0002, no emission from the project activity was identified, and PE_y = 0.

5.3 Leakage

According to the methodology ACM0002, the project leakage is not considered, and therefore, L_y=0.

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals(tCO _{2e})	Project emissions or removals (tCO _{2e})	Leakage emissions (tCO _{2e})	Net GHG emission reductions or removals(tCO _{2e})
28/09/2012-31/12/2012	26,556	0	0	26,556
01/01/2013-31/12/2013	101,189	0	0	101,189
01/01/2014–31/12/2014	87,394	0	0	87,394
01/01/2015–31/12/2015	84,343	0	0	84,343
01/01/2016–31/12/2016	90,719	0	0	90,719
01/01/2017–20/12/2017	82,496	0	0	82,496
Total	472,697	0	0	472,697

During the monitoring period, the actual emission reduction of 1910 days is 472,697tCO_{2e}, which is 0.02% lower than the estimated emission reduction⁷ of 482,382tCO_{2e}, which will not cause the over estimation of emission reductions. The deviation is within a reasonable range.

⁷ <https://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view>

APPENDIX 1: <EVIDENCE OF ACHIVED SD CONTRIBUTIONS >

SDG Target 7.2, SDG Indicator 7.2.1 Renewable energy share in the total final energy consumption

Period	Amount of renewable energy (MWh)	Data source
08/11/2011-27/09/2012	88,417.560	https://registry.verra.org/app/projectDetail/VCS/969
28/09/2012-20/12/2017 (This monitoring period)	507,789.778	Section 5 of this report
Current Project Contributions	507,789.778	Only the amount of renewable energy supplied to the Grid by the project during this monitoring period is used to calculate the current project contributions.
Contributions Over Project Lifetime	596,207.338	Contributions over project lifetime to SDGs 7 include the amount of renewable energy supplied to the Grid by the project in all previously approved VCS monitoring reports and the amount during this monitoring period.

SDG Target 8.3, SDG Indicator 8.3.1 Proportion of informal employment in non-agriculture employment, by sex



华润电力风能（青岛）有限公司-平度隆鑫风电场
员工名单（2017 年）

序号	姓名	职位	性别
1	孙松磊	场长	男
2	张光明	安全管理岗	男
3	腾鹏	检修巡操岗	男
4	邓健	检修巡操岗	男
5	罗鹏	检修巡操岗	男
6	唐雪海	检修巡操岗	男
7	王锋	检修巡操岗	男
8	刘衍	检修巡操岗	男
9	安绍璞	检修巡操岗	男
10	白宇	检修巡操岗	男
11	陈强波	检修巡操岗	男
12	丛健	行政后勤岗	男
13	蔡至诚	检修巡操岗	男
14	施朋	检修巡操岗	男
15	王超	检修巡操岗	男
16	张腾辉	检修巡操岗	男
17	张书峰	检修巡操岗	男
18	李佳卓	检修巡操岗	男
17	戴薇薇	会计（兼职）	女
18	苏毅	行政后勤岗（兼职）	女
19	丁文雁	档案管理岗	女
20	谢珍珍	行政后勤岗	女
21	白珊珊	人力资源岗（兼职）	女
22	韩芳	商务岗（兼职）	女

SDG Target 13, SDG Indicator Tonnes of greenhouse gas emissions avoided or removed

Period	Amount of emission reductions (tCO₂e)	Data source
08/11/2011-27/09/2012	82,302	https://registry.terra.org/app/projectDetail/VCS/969
28/09/2012-20/12/2017 (This monitoring period)	472,697	Section 5 of this report.
Current Project Contributions	472,697	Only the amount of emission reductions during this monitoring period is used to calculate the current project contributions.
Contributions Over Project Lifetime	554,999	Contributions over project lifetime to SDGs 13 include the amount of emission reductions avoided release into the atmosphere by the project in all previously approved VCS monitoring reports and the amount during this monitoring period.